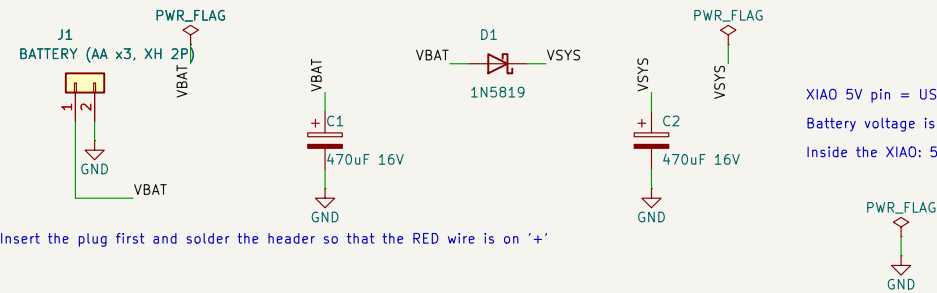


Linetracer2 Lite rev.L3 : XIAO ESP32C6 on pin headers, 3 x LBR–127HLD, 1 button, 1 LED, 1 buzzer

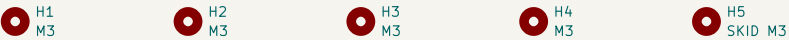
Power: AA x3 (alkaline 4.5V recommended / NiMH 3.6V) -> VBAT (motors) ; VBAT -> D1 Schottky -> VSYS -> XIAO 5V pin. USB 5V can not back-feed the batteries. Never use 4 cells (the XIAO's 5V input clamps above about 6 V).

1. POWER

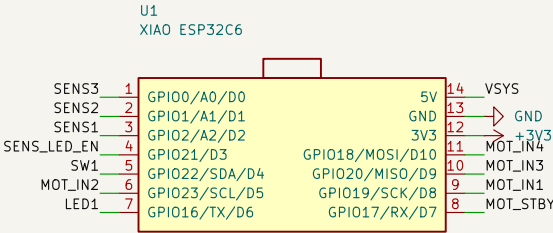


Insert the plug first and solder the header so that the RED wire is on '+'

MECHANICAL: H1–H4 = gearbox frame M3 (x30/x70, y62/y96), H5 = front ball caster / skid, snap-in (x50, y11: behind the centre sensor)

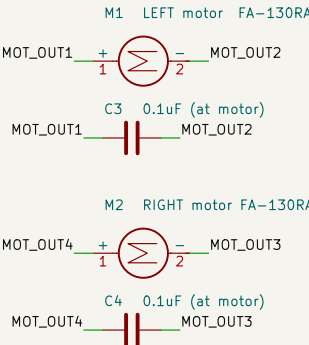
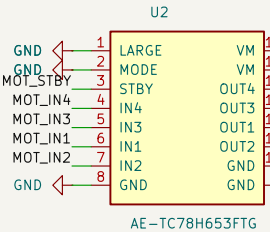


2. MCU Seeed Studio XIAO ESP32C6 (on two 1x7 pin headers)



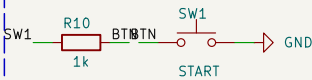
GPIO15 = yellow user LED on the XIAO (on = 0), a 2nd indicator for free. BOOT / RESET buttons are on the XIAO.
Reset state (ESP32–C6 datasheet): GPIO18–23 weak pull-up during reset, then floating;
GPIO16 (TX) / GPIO17 (RX) pull-up after reset + boot log / UART REPL -> no button and no motor input there.
On pin headers the XIAO sits 2.5 mm above the board: its bare bottom test pads can not touch the tracks.
Never solder a Li–ion cell to the XIAO's BAT pads on this robot (AA cells are on the 5V pin).

3. MOTOR DRIVER (Akizuki AE–TC78H653FTG module, DIP16)

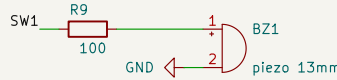


LARGE=L : small mode (2 motors, 2A/ch) MODE=L : IN/IN control
IN1/IN2 -> OUT1/OUT2 = LEFT motor, IN3/IN4 -> OUT3/OUT4 = RIGHT motor (+ = OUT4)
H/L fwd, L/H rev, H/H brake, L/L coast. Over–current (ISD) latches OFF:
toggle STBY L->H to recover (done by the library).

4. BUTTON / LED / BUZZER



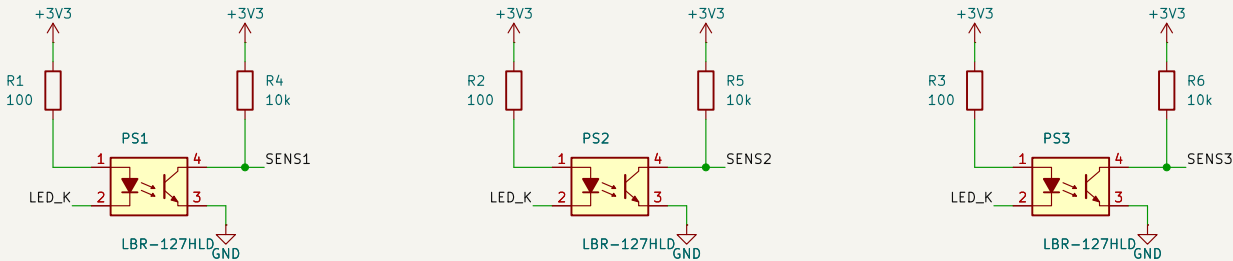
internal pull-up, pressed = 0. R10: if START is pressed while D4 beeps, only 3.3 mA flows
short press = start / stop, long press = speed level



The buzzer shares D4 with the START button (no free XIAO pin). A piezo passes no DC, so reading the button (input + pull-up) works: to beep, D4 becomes a PWM output – it also sounds while START is held.

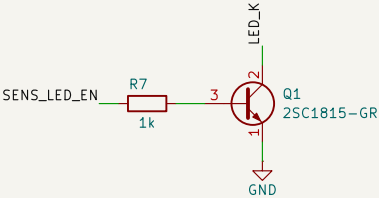
5. LINE SENSORS 3 x LBR–127HLD (mount on the BOTTOM side, lens to the floor), 12 mm apart

IR LED: (3.3V – 1.2V – 0.1V) / 100 ohm = about 20 mA each. Q1 switches all LEDs (ambient light cancel). Photo transistor: 10k pull-up -> white = low voltage, black = high voltage
The body is 5.6 mm tall: the front of the robot is lifted by the 7.5 mm ball caster or skid (lens about 2.25 mm above the floor)



PS1 = LEFT -> D2 (GPIO2/ADC), PS2 = CENTRE -> D1 (GPIO1/ADC), PS3 = RIGHT -> D0 (GPIO0/ADC) (seen from above, robot moving up the page)

Q1: all IR LEDs on when SENS_LED_EN = 1 (about 60 mA)



Design notes: lite/README.md
XIAO ESP32C6 + Akizuki AE–TC78H653FTG module + 3 x LBR–127HLD
All parts through–hole. XIAO on two 1x7 pin headers

EDTC
Sheet: /
File: Linetracer2–Lite.kicad_sch
Title: Linetracer2 Lite rev.L3 – XIAO ESP32C6 kids line tracer

Size: A3 Date: 2026–10–07 Rev: L3
KiCad E.D.A. 10.0.6 Id: 1/1